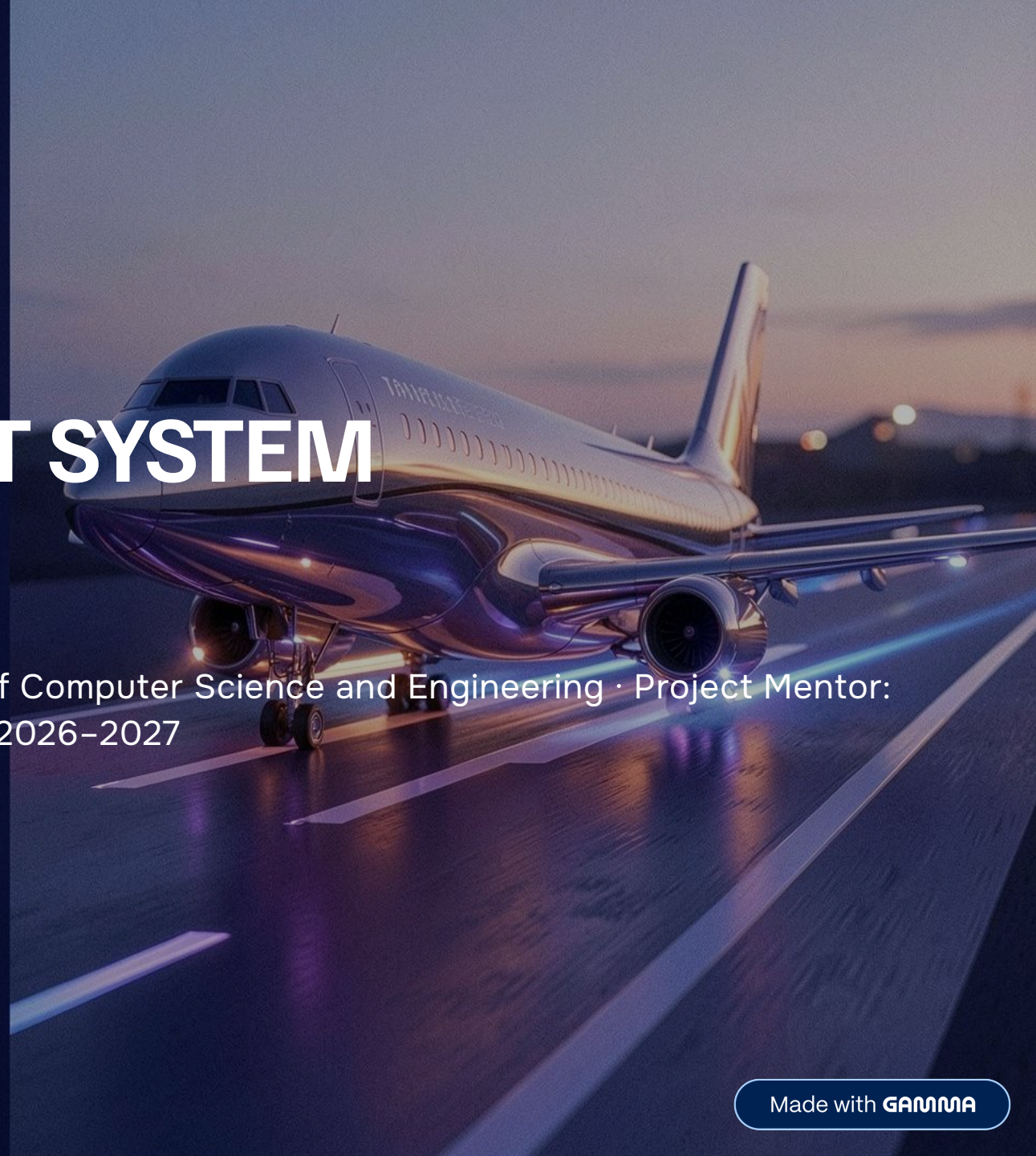
AIRLINE MANAGEMENT SYSTEM

Airline Reservation and Passenger Management System

Arya College of Engineering & IT, Jaipur · Department of Computer Science and Engineering · Project Mentor: Er. Ram Babu Buri · Team SkyTech · Academic Session: 2026–2027



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02

Manual reservation processes create fragmented records

- SkyTech Airlines is proposed as a centralized Java-based web application for flight search, passenger information, bookings, tickets, cancellations, and administrative operations
- Problems addressed: manual booking and passenger records; duplicate or inconsistent data; difficult flight and schedule searches; booking and cancellation errors; limited administrative visibility

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03

Five modules turn reservation tasks into one connected system

Connected responsibilities across the reservation journey

Passenger registration, authentication, profile management, and role-based access

Flight search by origin, destination, and date with schedules, fares, and database-derived availability

Passenger details, booking confirmation, unique PNR generation, ticket retrieval, booking history, and cancellation

Protected administrator dashboard for flights, routes, schedules, fares, bookings, and reports

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04

The proposed stack separates presentation, logic, and data access

All technologies are proposed; local installation and implementation status remain to be verified.

01

Java 21 · JSP · HTML · CSS · JavaScript · Jakarta Servlets · JDBC · MySQL

02

Apache Tomcat 10.1 · Apache Maven · Git and GitHub

03

Browser → JSP Presentation Layer → Servlet Controller Layer → Service Layer → DAO/JDBC Layer → MySQL Database

04

Entities: Users, Flights, Bookings, Passengers, and Tickets

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05

Five owners, one integrated project

Module dependencies connect authentication, flight data, booking, ticketing, cancellation, and administration.



Authentication and User Management

Mohammed Sahil



Flight Search and Availability

Nikhil Ranjan Choubey



Booking and Passenger Management

Mukesh Sah



Ticketing and Cancellation

Muskan Raj



Admin Dashboard and Flight Management

Prashant Chaudhary

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06

Booking is validated before it is committed

Login → Search Flights → Select Flight → Enter Passenger Details

Validate availability → Generate unique PNR → Save booking

Booking confirmation → View ticket

Controls before commit

- Server-side validation and availability rechecking
- Unique PNR generation with a database uniqueness constraint
- Database transactions and duplicate-submission prevention

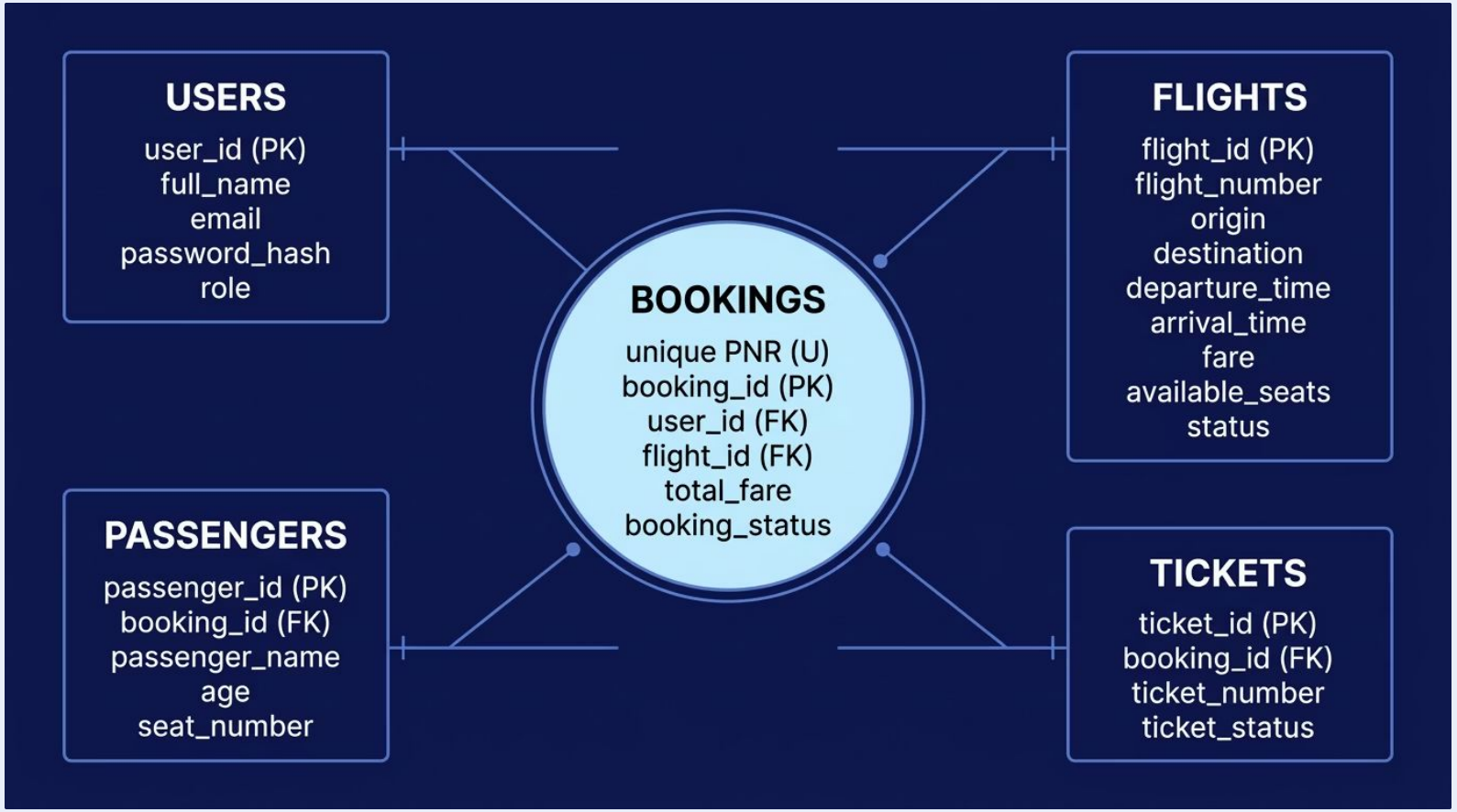
The baseline excludes verified payment processing and live airline inventory.

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07

The proposed database preserves traceability

Show primary keys, foreign keys, unique PNR, and relationships between users, flights, bookings, passengers, and tickets. Label this proposed conceptual database design.



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08

Security and reliability are requirements to verify



Salted password hashing, role-based access control, server-side validation, prepared SQL statements, database constraints, and transactions



Booking ownership checks, controlled error messages, session security, output encoding, and CSRF protection



Testing scope: unit, integration, functional, regression, security, and acceptance testing



Examples: invalid login, duplicate booking submission, PNR uniqueness, unauthorized admin access, and transaction rollback

The SRS defines 22 functional and 7 security test cases; none has been executed, so no pass/fail result is claimed.

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09

Collaboration supports the five-module build

Feature Branch → Development → Commit and Push

Pull Request → Merge into Main → Integration Testing

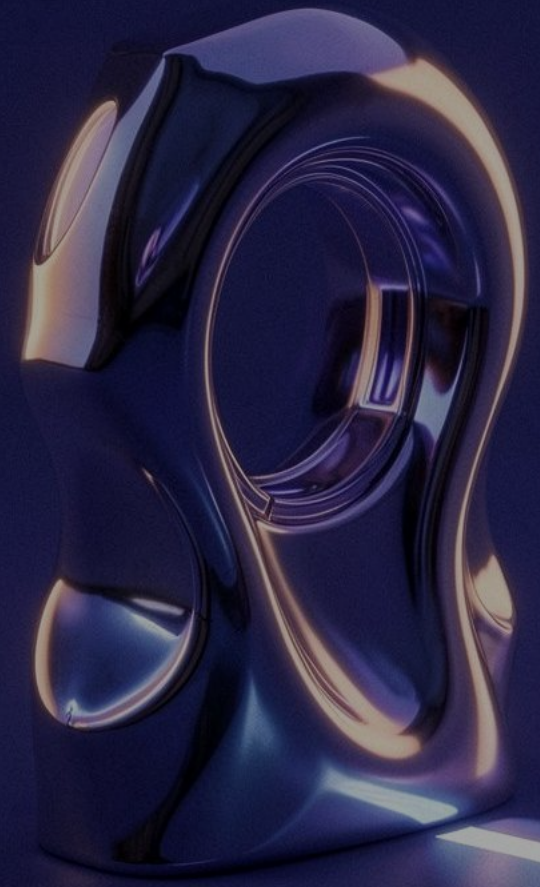


Limitations: no live airline inventory integration; payment processing and refunds outside the baseline; email/SMS delivery and production deployment not verified



Future enhancements: payment gateway integration; PDF tickets and notifications; graphical seat selection; external airline APIs; automated regression and load testing

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AIRLINE MANAGEMENT SYSTEM

SkyTech Airlines proposes a modular Java web application for reservation activities, supported by database integrity, maintainable layers, team collaboration, and explicit verification of implementation and testing.

THANK YOU

Questions & Answers

Team SkyTech — Arya College of Engineering & IT

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